



Life without tectonics

Research has indicated that plate tectonics on exoplanets are not a requirement to support life
<https://digit.in/sep18-72-1>



Whale sonar

A new paper hypothesizes that whales use their songs as a kind of sonar to identify and home in on mates. <https://digit.in/sep18-72-2>

Fighting mosquitoes with tech

Mosquitoes are more deadly to humans than any other animal. Now, researchers are using drones, robotics and gene editing to tackle the problem.

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The Anopheles mosquito

According to the World Health Organisation, in the year 2015 alone, malaria alone was responsible for 438,000 deaths around the world. The Aedes, Anopheles and Culex mosquitoes are responsible for the spread of various diseases, such as chikungunya, Zika, malaria, dengue, yellow fever and the West Nile virus. More countries are reporting outbreaks of these diseases, and the number of people affected is increasing every

year. Humans are now fighting back, with the latest technologies.

DRONES

Adult mosquitoes have few priorities in life. Once they have fed and rested, they typically go to mate, which occurs around the breeding areas. One of the most important aspects of controlling mosquitoes is identifying the breeding areas. The problem is that the mosquito populations and breeding areas are constantly shifting. They can change even on a daily basis because of the

weather conditions. Drones can survey large areas in little time, and identify the mosquito breeding “hotspots” – pools of water, culverts, irrigated fields or open containers.

Microsoft’s Project Premonition tackles the problem of mosquitoes, and one component of the initiative is to use drones to identify the breeding hotspots. The drones were tested in Grenada. Microsoft is developing technologies to automate the process, to make sure that the drone can identify a flight path, and not fail during the flight. There are plans to develop drones that can directly capture the mosquitoes as well – to sample and check them for diseases.

Researchers from the Aberystwyth University have also used a similar approach to identify breeding hotspots in Zanzibar. In the rural and agricultural areas of Zanzibar, there are many small nooks and crannies where mosquitoes can breed. Identifying all of them on foot is a long and laborious process. A basic off the shelf DJI drone can be effectively used to survey 30 hectares of rice paddy in a single twenty minute run. Once identified, the breeding spots can be sprayed with pesticides to kill the larvae. A typical larvicide campaign can be wrapped up in under a day. The program has been so successful, that in some areas of Zanzibar, the prevalence of malaria has dropped from 40 percent of the population to less than 1 percent.

Drones are also used to fight mosquitoes using another innovative approach. Genetically modified sterile mosquitoes are introduced to the wild, through the use of drones. Essentially, the drones act as dropships for the armies of mosquitoes that fight for humans. WeRobotics used this approach in Brazil to contain the spread of Zika.

MODIFIED MOSQUITOES

In 1998, UC Irvine biologist Anthony James developed the genetically modified mosquito model, the first to do so anywhere in the world. Since then, the focus of his work is to use genetically modified mosquitoes